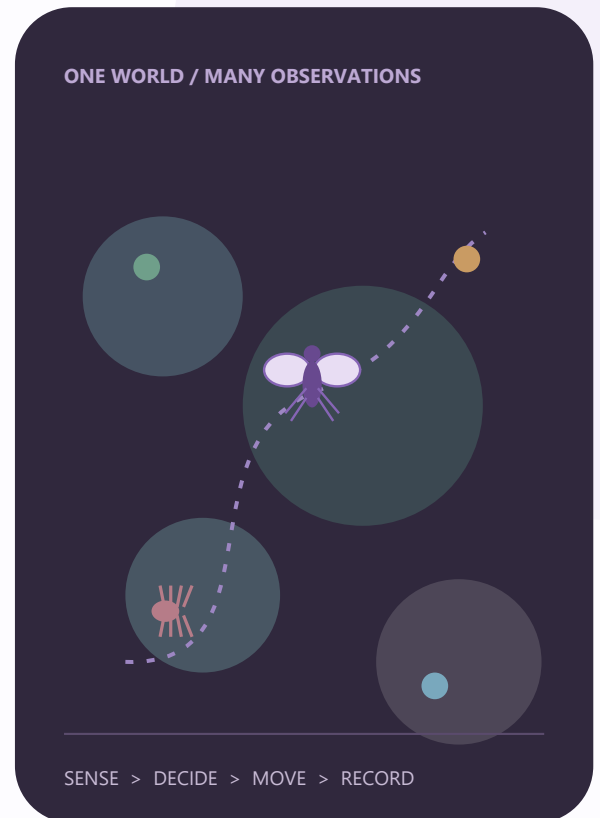


BRAIN RESEARCH, MADE EXPLORABLE

MOSK

A small life, unfolding.

What happens when a published brain map becomes a question you can test on screen? MOSK is an interactive observatory: follow one virtual fruit fly, change its world, and inspect the signals behind each visible response.



THE PRODUCT IN THREE MOVES

01 OBSERVE

Watch the fly explore a world that unfolds around it.

02 INTERVENE

Place food, water or a spider and compare what changes.

03 EXPLAIN

Read controller signals, life events and real source annotations.

A clear scientific boundary. The fly on screen is an authored simulation. Google Research and collaborators published anatomical wiring data, not a ready-made virtual animal. MOSK makes that research approachable while preserving the distinction.

The research beneath the experience

Three related layers serve different purposes. Their provenance is visible throughout MOSK.

PUBLISHED / MALECNS

Google and collaborators mapped the male fly's central nervous system.

The Cell study describes **166,691 neurons** across brain and ventral nerve cord, with roughly **125 million synaptic contacts**. The released structural map supports tracing sensory-to-motor paths and comparing cell types; it is not a recording of a living fly's activity.

IN MOSK / SOURCE ATLAS

A real anatomical sample you can inspect.

The MaleCNS atlas contains **96 source-identified annotation records** with cell labels, classes and provenance. Visitors can search and inspect them. It does not bundle the full connection graph or make those cells control the roaming fly.

THREE LAYERS, CLEARLY LABELED

01

Anatomy

Real MaleCNS identifiers and cell annotations.

02

Roaming

A designed, interpretable 18-channel controller.

03

Memory

A separate published mushroom-body assay.

MEMORY ASSAY / INDEPENDENT STUDY

A second published model makes learning observable without pretending that the MaleCNS map already supplies it.

The Research view runs the Huang-Luo Nature 2024 mushroom-body model: odor conditioning, retention and extinction change modeled synaptic weights and circuit response rates. Its checkpoint belongs to a life record, but those weights **do not steer the roaming fly**. The assay is not part of Google's MaleCNS release.

Watch a decision take shape

The point is to test a cue, then follow its effect through body, behavior and record.

01 / THE LIVING WORLD

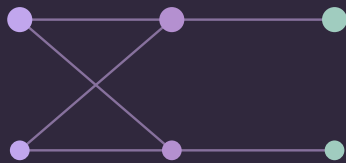
Explore an endless, local habitat.

The fly reveals procedurally generated terrain as it moves. The map shows its path, food, water, spiders and protective cover. Pan, zoom, follow, pause or change speed. Use the three brushes to add a stimulus to explored terrain and watch what happens.

Six live measures - energy, hunger, hydration, fatigue, health and arousal - make the cost of each choice visible.



SENSORY DRIVE MOTOR



Illustrated controller channels

02 / BRAIN ACTIVITY

Inspect why the model acted.

Hover, focus or pin one of **18 channels** to see its current activation, input and illustrated influences. If an exposed fly detects a nearby spider, the modeled Threat and Escape channels rise; the controller changes forward and turning commands and the fly flees.

These are normalized controller values, **not measured firing of MaleCNS neurons**.

A CONTROLLED OBSERVATION

PLACE

Spider nearby



SENSE

Threat rises



ACT

Escape turn



RECORD

Journal event

The observer can compare this event with a matched run or another generation.

A research notebook, not just a game

The app keeps observations, explains its sources and lets visitors test their own questions.

03 / EXPERIMENT JOURNAL

Keep the story of each life.

A timeline records birth, resources, threats, interventions, exploration and death. Charts and metrics summarize the life; archived generations remain available for comparison. Saved memory-assay checkpoints can be inspected read-only.

04 / RESEARCH & LEARNING

Go back to the evidence.

Run the separate odor-memory assay and inspect changing model weights. Search the MaleCNS reference atlas by source-identified cells. The in-app tutorial and About panel explain what each view means and where the science ends.

TECHNICAL FOUNDATION

React + TypeScript + Vite power a seeded, deterministic 30 Hz simulation. The world generates near the fly rather than building infinity in advance. Local sensory input drives an interpretable utility controller; the separate published memory assay has its own circuit state. IndexedDB autosaves lives, while JSON import/export provides portable backups. Naming and new generations support controlled comparisons; no learned state is inherited.

Explore the demo and original sources

Live MOSK demo - moskdemo.netlify.app

Google Research overview of the MaleCNS release

Berg et al., Cell 2026 - doi.org/10.1016/j.cell.2026.08.015

Official dataset and provenance - male-cns.janelia.org

Huang, Luo et al., Nature 2024 - doi.org/10.1038/s41586-024-07819-w

MOSK source code - github.com/francopignanelli/mosk

Made by **Franco Pignanelli** as an independent visual and interactive exploration of published work. MOSK is not affiliated with Google or HHMI Janelia.